Divisors on Curves: Zeros, Poles, and Riemann–Roch

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On a compact connected Riemann surface X, a nonzero meromorphic function f has an isolated integer order at every point. In a local coordinate $z(p) = 0$,

$$f(z) = z^m u(z), \qquad u(0) \neq 0, \qquad \operatorname{ord}_p(f) = m.$$

A change of coordinate multiplies z by a holomorphic unit, so m is intrinsic. The basic treatment here follows Forster, *Lectures on Riemann Surfaces*, and the definitions and degree results of the Stacks Project.[1]

§1.1 A divisor remembers every local condition

A divisor is a finite integral sum $D = \sum_p n_p[p]$. Its degree is $\deg D = \sum_p n_p$. The principal divisor of a nonzero meromorphic function is

$$(f) = \sum_p \operatorname{ord}_p(f)[p], \qquad (fg) = (f) + (g).$$

On the Riemann sphere $\mathbb{P}^1$, the coordinate $w = 1/z$ near infinity gives

$$z - a = w^{-1}(1 - aw), \qquad (z - a) = [a] - [\infty].$$

In particular, the affine zero at a must be balanced by a pole at infinity.

Theorem 1.1.1 (Degree zero)
For a nonzero meromorphic function on a compact connected Riemann surface, $\deg(f) = 0$.

For a nonconstant f, the induced holomorphic map $X \to \mathbb{P}^1$ has the same degree when counted above 0 or above ∞. The respective local multiplicities are exactly the orders of its zeros and poles. For constant $f \neq 0$, the divisor is zero.

§1.2 Allowed poles form a vector space

For divisors write $D \geq 0$ when each coefficient is nonnegative. Set

$$L(D) = \{0\} \cup \{f \neq 0 \text{ meromorphic} : (f) + D \geq 0\}, \qquad \ell(D) = \dim_{\mathbb{C}} L(D).$$

With $D = n[\infty]$ on $\mathbb{P}^1$, the condition forbids finite poles and permits a pole of order at most n at infinity. Such rational functions are exactly polynomials of degree at most n:

$$L(n[\infty]) = \operatorname{span}\{1, z, \dots, z^n\}, \qquad \ell(n[\infty]) = n + 1 \quad (n \geq 0).$$

Prescribing a zero at 0 changes the divisor to $2[\infty] - [0]$, and gives

$$L(2[\infty] - [0]) = \operatorname{span}\{z, z^2\}.$$

[1] Stacks Project, Tag 02AQ, with related material in Tag 02RS.

Adding a further zero at 1 leaves just $\mathbb{C}z(z-1)$. This makes the bookkeeping precise: a pole allowance raises the possible degree, while a vanishing condition imposes a linear constraint.

Proposition 1.2.1

If $\deg D < 0$, then $L(D) = \{0\}$.

Proof. If $f \neq 0$ belonged to $L(D)$, the effective divisor $(f) + D$ would have nonnegative degree, whereas $\deg((f) + D) = \deg(f) + \deg D = \deg D < 0$, a contradiction. □

§1.3 Line bundles explain the local-to-global problem

On a small open set U, choose a meromorphic function g_U with orders prescribed by $D|_U$. The allowable local functions are $g_U^{-1}\mathcal{O}_X(U)$. On overlapping charts U, V, the ratio g_U/g_V is holomorphic and nowhere zero. Those ratios are precisely transition functions for a holomorphic line bundle $\mathcal{O}(D)$. Thus

$$L(D) = H^0(X, \mathcal{O}(D)).$$

The vector space is finite-dimensional for compact X. Divisors turn local restrictions on poles into globally compatible sections; the obstruction is that independently chosen local sections need not glue.

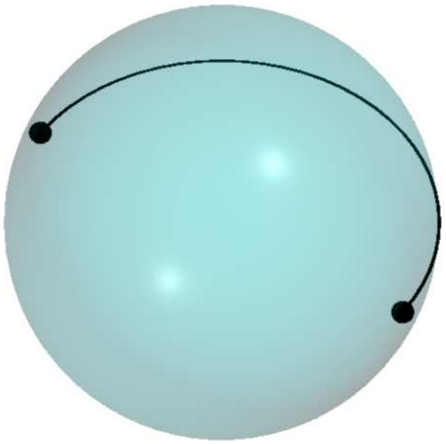


Figure 1.1: Local orders of functions on a compact curve are constrained by global analytic and topological data.

§1.4 Canonical divisors and the genus

A meromorphic differential is locally $h(z)\,dz$. Changing to another local coordinate multiplies the coefficient by a nonzero holomorphic factor, so $\operatorname{ord}_p(h(z)\,dz)$ is well defined. The divisor $K = (\omega)$ of a nonzero meromorphic differential is a canonical divisor. Different choices of ω differ by a meromorphic function, hence determine linearly equivalent divisors.

On $\mathbb{P}^1$ the calculation

$$dz = -w^{-2}\,dw, \qquad w = 1/z,$$

shows $K = -2[\infty]$ and $\deg K = -2$. A complex torus $\mathbb{C}/\Lambda$, by contrast, carries the nowhere-vanishing holomorphic differential dz, so $K = 0$. The general degree formula is

$$\deg K = 2g - 2$$

for a compact Riemann surface of genus g. This is a global theorem; the sphere and torus are tests of its normalization, not its proof.

§1.5 Riemann–Roch counts sections and obstructions

Theorem 1.5.1 (Riemann–Roch)
If D is a divisor on a compact connected Riemann surface of genus g, then

$$\ell(D) - \ell(K - D) = \deg D + 1 - g.$$

We quote the general theorem rather than disguise its cohomological proof as a calculation. The correction $\ell(K - D)$ counts meromorphic differentials whose divisor is at least D; these are the global constraints missing from a naive dimension count.

Take $X = \mathbb{P}^1$, $D = n[\infty]$ with $n \geq 0$, and $K = -2[\infty]$. Since $\deg(K-D) = -2-n < 0$, the preceding proposition gives $\ell(K - D) = 0$. Hence

$$\ell(D) = n + 1,$$

recovering the explicit polynomial basis. The theorem has passed a nontrivial consistency check.

On an elliptic curve $g = 1$, $K \sim 0$. For a divisor of positive degree d, $\deg(-D) < 0$ gives $\ell(-D) = 0$, and so $\ell(D) = d$. In particular

$$\ell([p]) = 1, \qquad \ell(2[p]) = 2.$$

The first equality says that a nonconstant meromorphic function cannot have just one simple pole. The second says a function with at most one double pole does exist. This is the first dimension-count explanation for why elliptic functions naturally develop double poles.

§1.6 Ramification is another kind of local multiplicity

For a nonconstant holomorphic map $F : X \to Y$ of degree d, a local coordinate expression at p has the form $z \mapsto z^{e_p}$ after suitable changes of coordinates. The Riemann–Hurwitz theorem states

$$2g_X - 2 = d(2g_Y - 2) + \sum_{p \in X} (e_p - 1).$$

For the rational map $F(z) = z^d$ on $\mathbb{P}^1$, only 0 and ∞ ramify, each with index d. The right-hand side is

$$-2d + (d - 1) + (d - 1) = -2,$$

as required for a sphere.

Riemann–Roch and Riemann–Hurwitz answer different questions. One counts sections subject to allowed zeros and poles; the other compares the canonical divisors of curves connected by a map. Their common arithmetic is the addition of local integer orders to obtain a global constraint.